KinCare: Data Set, Schema and Data Types

IT for Managers Capstone · Section 4 · Group 5 (TCS)

1. The data set

KinCare stores data about **people** (families, parents, caregivers), **bookings** (who cares for whom, and when) and **proof of care** (each visit, plus alerts when something goes wrong). It is held in **7 tables** in Supabase. Each table has one ID that identifies every row, and tables link to each other through these IDs.

2. Data elements and data types

PK = primary key (the row's own ID). FK = foreign key (links to another table's ID).

USERS everyone who logs in

Element	Data type
user_id (PK)	UUID
name	Text
email	Text
phone	Text
role	Text (family / caregiver / elder / coordinator / admin)
created_at	Timestamp

PARENTS the elder receiving care

Element	Data type
parent_id (PK)	UUID
family_user_id (FK)	UUID → USERS
name	Text
age	Integer
address	Text
pin_code	Text (6 digits)
care_needs	Text
emergency_contact	Text
consent_given	Boolean (yes/no)
consent_at	Timestamp

CAREGIVERS verified attendants

Element	Data type
caregiver_id (PK)	UUID
user_id (FK)	UUID → USERS
photo_url	Text (link)
pin_code	Text (6 digits)
daily_rate	Decimal (₹)
id_checked	Boolean
police_checked	Boolean
training_done	Boolean

BOOKINGS a 14-day care block

Element	Data type
booking_id (PK)	UUID
parent_id (FK)	UUID → PARENTS
caregiver_id (FK)	UUID → CAREGIVERS
shift_hours	Integer (12 or 24)
start_date	Date
end_date	Date
total_amount	Decimal (₹)
status	Text (active / completed / cancelled)

SHIFTS one per day of a booking

Element	Data type
shift_id (PK)	UUID
booking_id (FK)	UUID → BOOKINGS
caregiver_id (FK)	UUID → CAREGIVERS
shift_date	Date
start_time	Timestamp
status	Text (scheduled / checked in / completed / missed)

VISITS proof of care for a shift

Element	Data type
visit_id (PK)	UUID
shift_id (FK)	UUID → SHIFTS
check_in_at / check_out_at	Timestamp
latitude / longitude	Decimal
photo_url	Text (link)
tasks_done	Text (list)
bp_systolic / bp_diastolic	Integer (mmHg)
sugar	Integer (mg/dL)

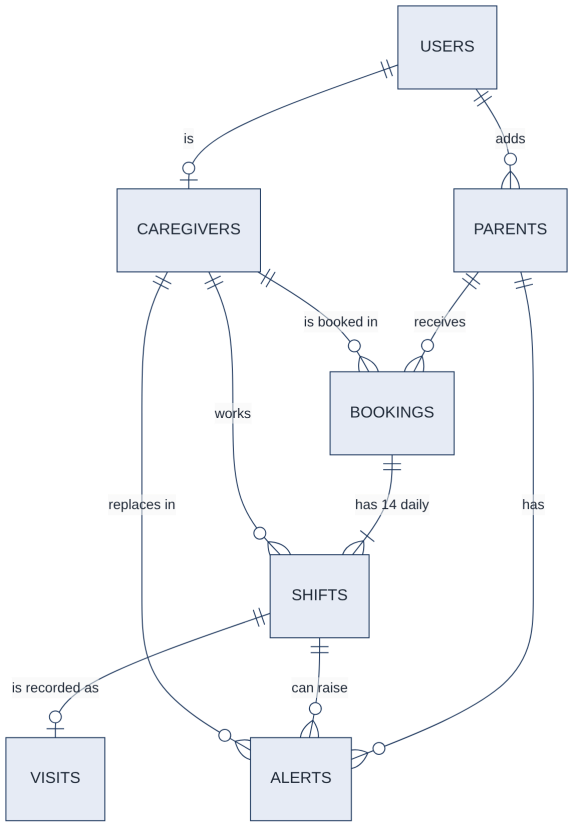
ALERTS no-show or SOS

Element	Data type
alert_id (PK)	UUID
parent_id (FK)	UUID → PARENTS
shift_id (FK)	UUID → SHIFTS (empty for SOS)
type	Text (no-show / SOS)
raised_at / resolved_at	Timestamp
replacement_caregiver_id (FK)	UUID → CAREGIVERS

Why these types: UUID gives every row a unique ID that can't be guessed. Pin codes and phone numbers are Text, not numbers, because we never calculate with them. Money and location are Decimal for exact values. Yes/no checks are Boolean. Dates are Date; exact moments (check-in, alerts) are Timestamp, so we can measure the 30-minute rule.

3. Relationships (the schema)

Relationship	Type	In plain words
USERS → PARENTS	One to many	One family member can add several parents.
USERS → CAREGIVERS	One to one	Each caregiver has one login.
PARENTS → BOOKINGS	One to many	A parent can have many bookings over time.
CAREGIVERS → BOOKINGS	One to many	A caregiver can be booked by many families.
BOOKINGS → SHIFTS	One to many	Each 14-day booking creates 14 daily shifts.
CAREGIVERS → SHIFTS	One to many	A caregiver works many shifts; a replacement takes over some.
SHIFTS → VISITS	One to one	Each shift has at most one visit record; a missed shift has none.
SHIFTS / PARENTS → ALERTS	One to many	A missed shift raises a no-show alert; a parent's SOS raises an SOS alert.
CAREGIVERS → ALERTS	One to many	A caregiver can be the replacement for many alerts.



Entity-relationship diagram. Two lines = exactly one; circle = zero allowed; crow's foot = many.